

1) Artificial Neural Network :

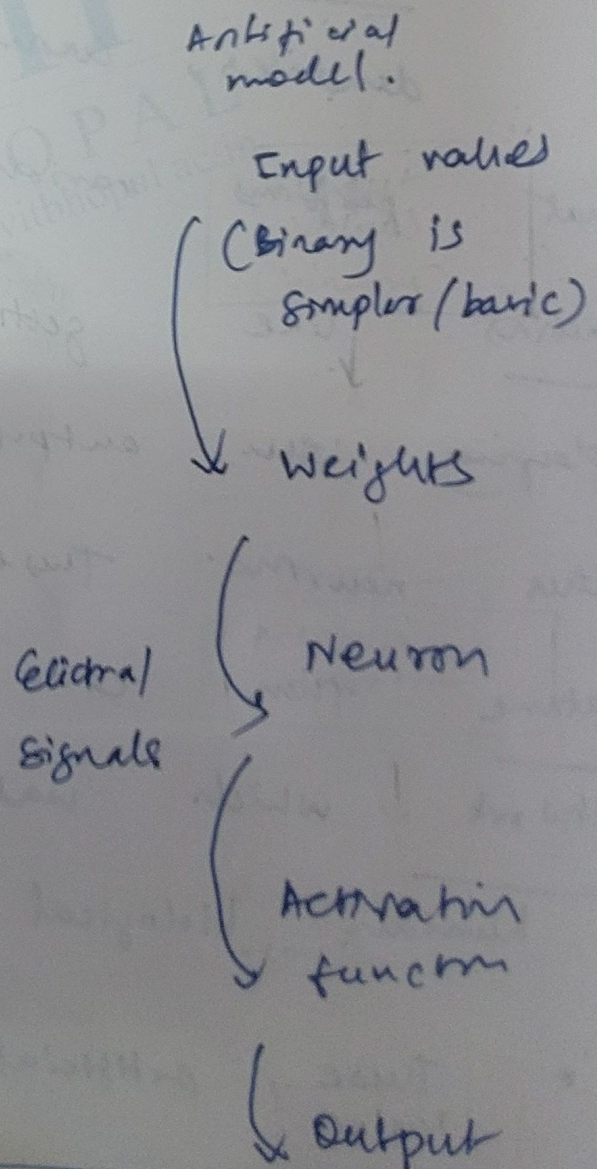
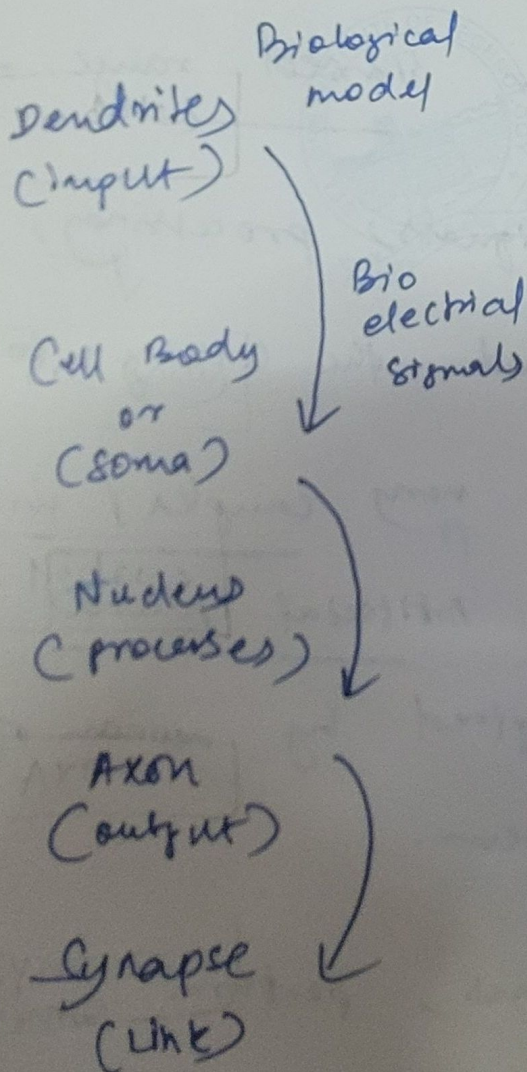
✓ on the contrary of the principal models of Artificial Neural Network, here are some of the key agendas in it.

✓ The ANN itself is a mimic of the biological neuron model which consists of dendrites, axons, nucleus, synapses that performs various or varied range of functions like getting signals, processing, displaying the outputs, and the linking to other neurons. These are very complex in nature. Then came the Artificial Neural network which was inspired by neuron of a human biological system.

✓ These artificial neurons, perform similar

kind of functions just like the biological neuron. It typically contains, input, a perception that processes the output, and an activation function that adds "non-linearity" to the output of the perception.

* Consider the given diagram for architectural comparison.



On the contrary of mathematical Basis and roles of key activation functions, these activation functions adds non-linearity to the summed value of the perceptron before deploying output.

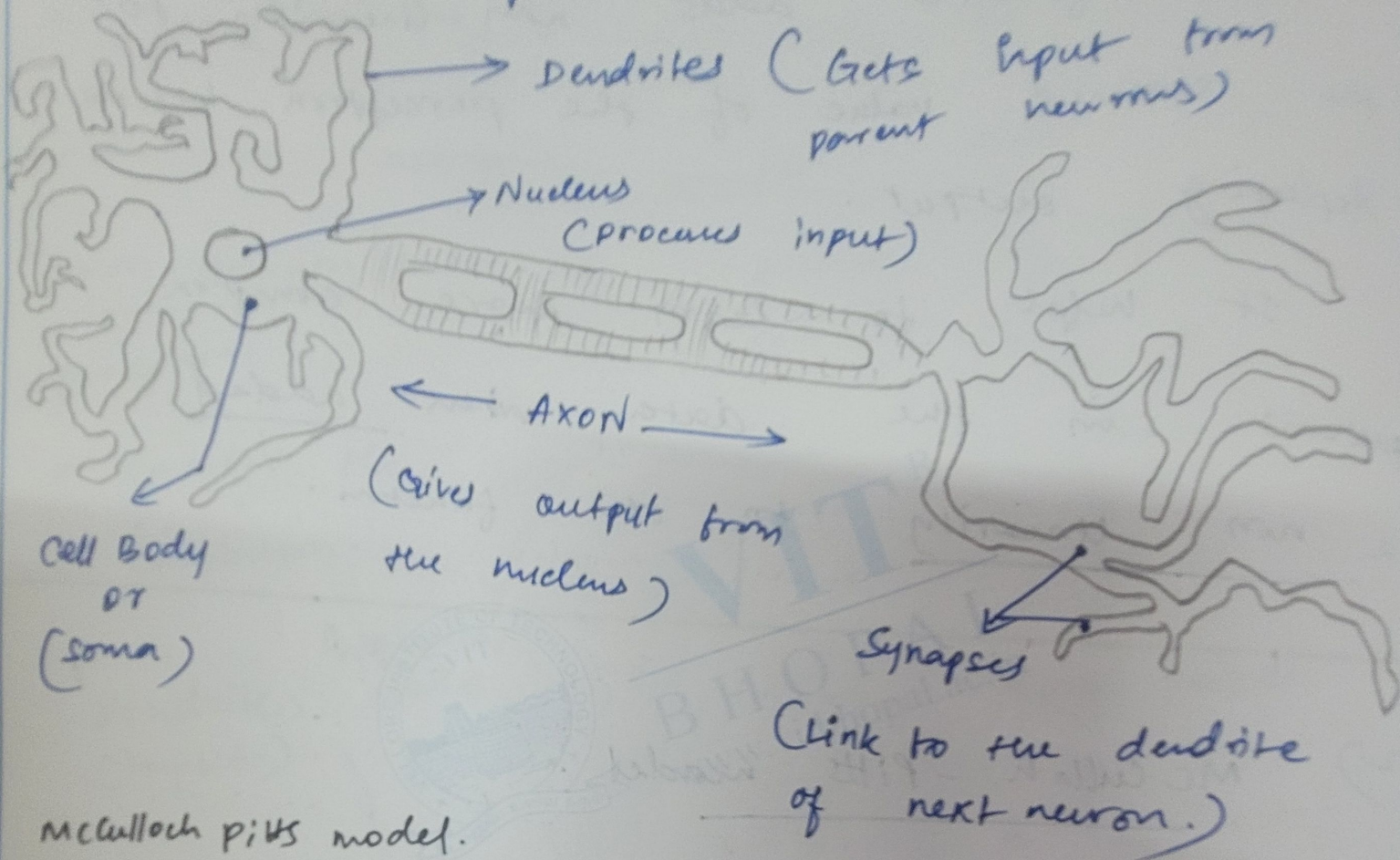
It helps to learn more complex patterns in the data when added a non linearity to the function.

2) McCulloch - Pitts model :

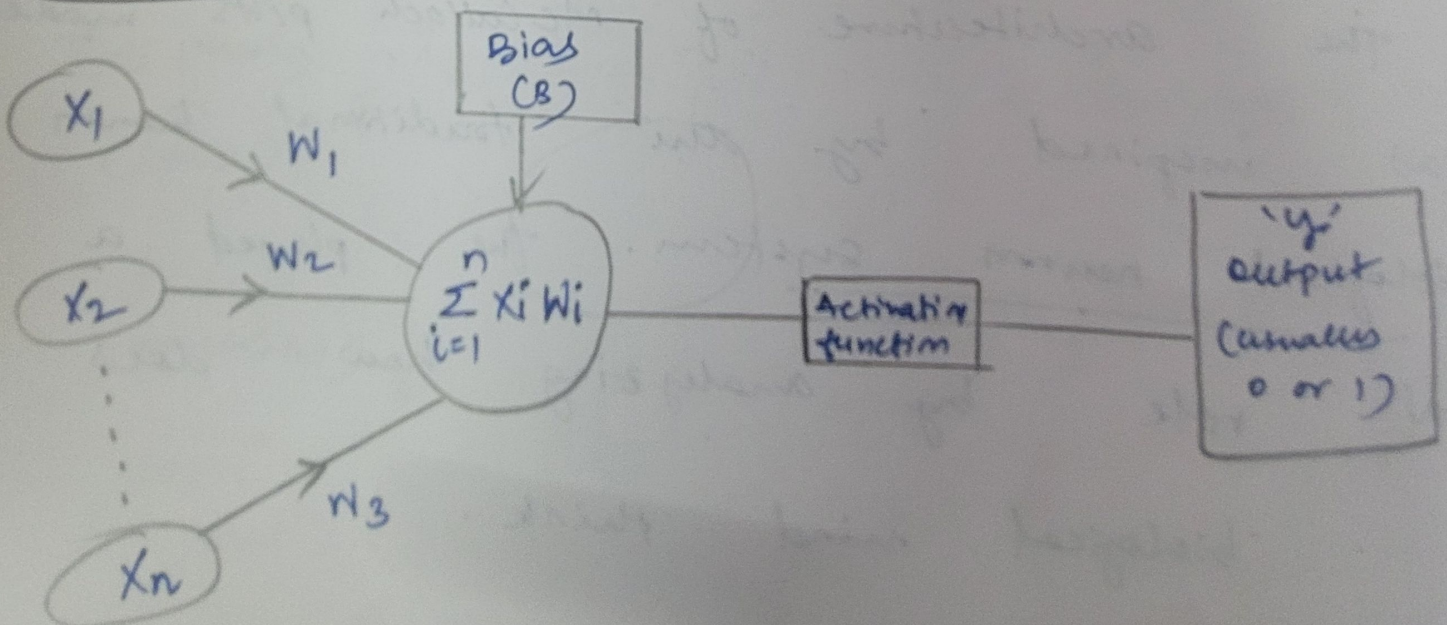
The architecture of McCulloch Pitts model was inspired by the traditional human biological neuron system. The played a vital role by analyzing how does a biological mind think.

★ As a result, a similar computable neuron model was created by the

neuron & pits. Here is a detailed comparison of the structure of neuron and McCulloch Pitts model.



McCulloch Pitts model.



W_n = weight of the input n

B = Bias

X_n = input of the neuron n

y = output

$$y = \sum_{n=1}^n X_i W_i + B$$

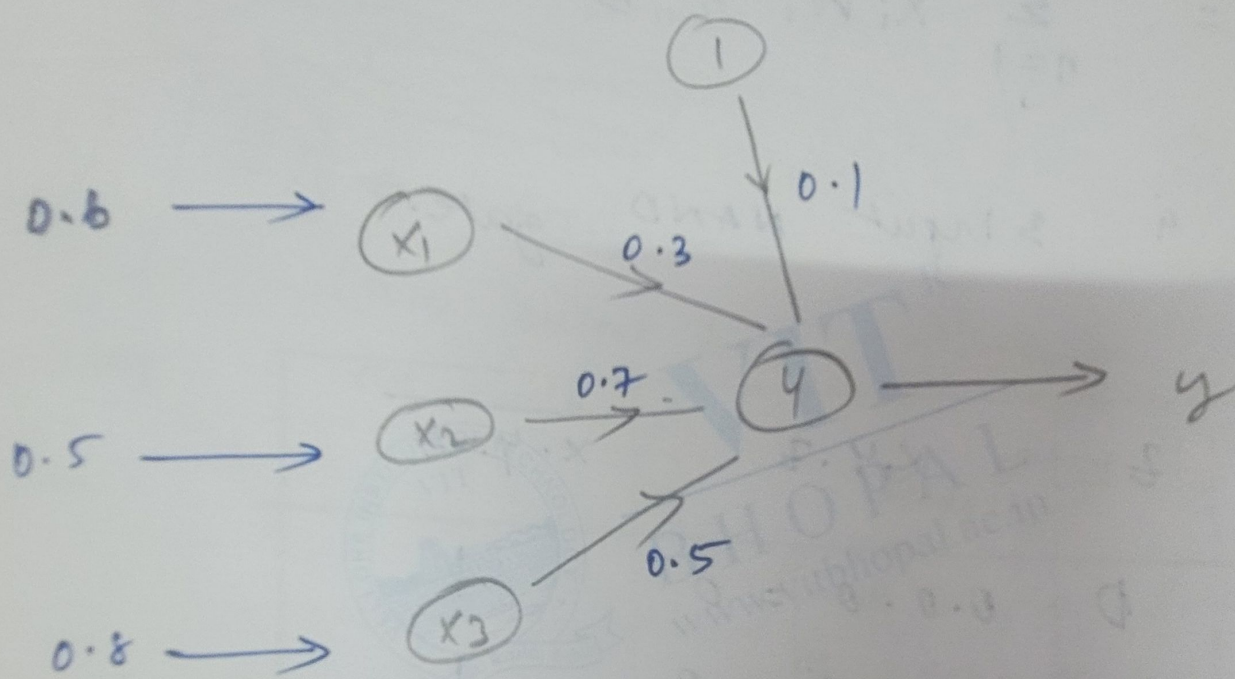
Here's a 3 input NAND gate.

x	y	z	$x \cdot y \cdot z$	$\overline{x \cdot y \cdot z}$
0	0	0	$0 \cdot 0 \cdot 0 = 0$	1
0	0	1	$0 \cdot 0 \cdot 1 = 0$	1
0	1	0	$0 \cdot 1 \cdot 0 = 0$	1
0	1	1	$0 \cdot 1 \cdot 1 = 0$	1
1	0	0	$1 \cdot 0 \cdot 0 = 0$	1
1	0	1	$1 \cdot 0 \cdot 1 = 0$	1
1	1	0	$1 \cdot 1 \cdot 0 = 0$	1
1	1	1	$1 \cdot 1 \cdot 1 = 1$	0

3) Output of the neuron 'y'.

⇒ Given that y is a neuron model.

Neuron structure:



a) Sigmoid function:

The function ranges from 0 to 1.

and the formula is given by

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \sum x_i w_i + b$$

$$z = (0.6)(0.3) + (0.9)(0.2) + (0.8)(0.5) + 0.1$$

$$z = 0.18 + 0.35 + 0.4 + 0.1$$

$$\boxed{z = 1.03}$$

Applying the value of z in $\sigma(z)$

$$\sigma(1.03) = \frac{1}{1 + e^{-1.03}}$$

$$= \frac{1}{1 + 0.357}$$

$$\therefore \boxed{e^{-1.03} \approx 0.3700696}$$

$$= \frac{1}{1.357}$$

$$\boxed{\sigma(z) = 0.736}$$

b) Tanh:-

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

The above is the formula to find the

tanh for the given neuron 'y'.

we know that $x = 1.03$;

$$\tanh = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

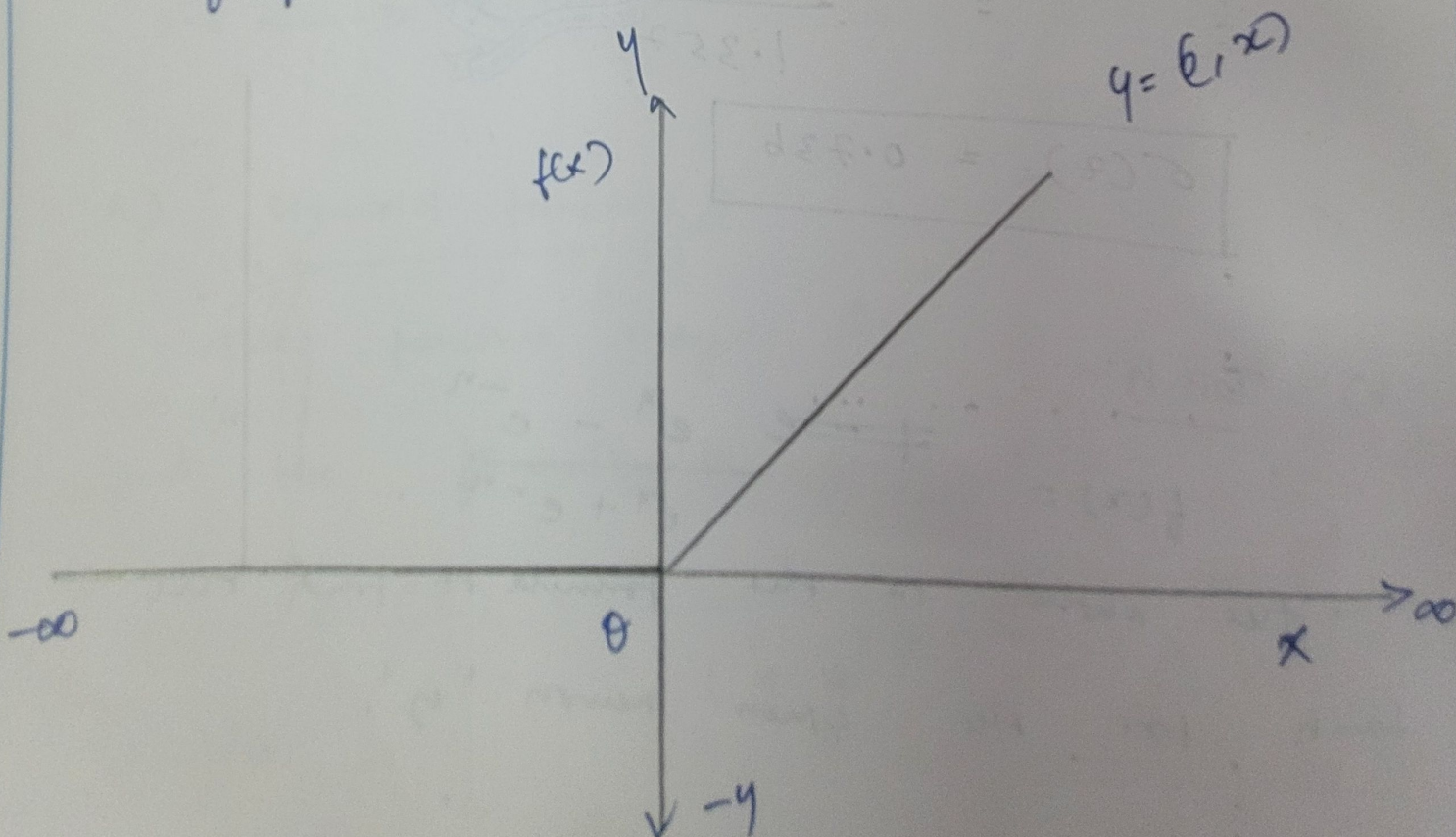
$$\begin{cases} e^{1.03} = 2.801 \\ e^{-1.03} = 0.357 \end{cases}$$

$$= \frac{e^{1.03} - e^{-1.03}}{e^{1.03} + e^{-1.03}} = \frac{2.801 - 0.357}{2.801 + 0.357}$$

$$= \frac{2.444}{3.158} \Rightarrow \boxed{0.773}$$

c) ReLU :

for ReLU, consider the given graph.



4) Activation Function :

* Activation functions in Neural Networks helps to add "non-linearity" to the function such that by doing this process, the model can learn very complex patterns from the given input.

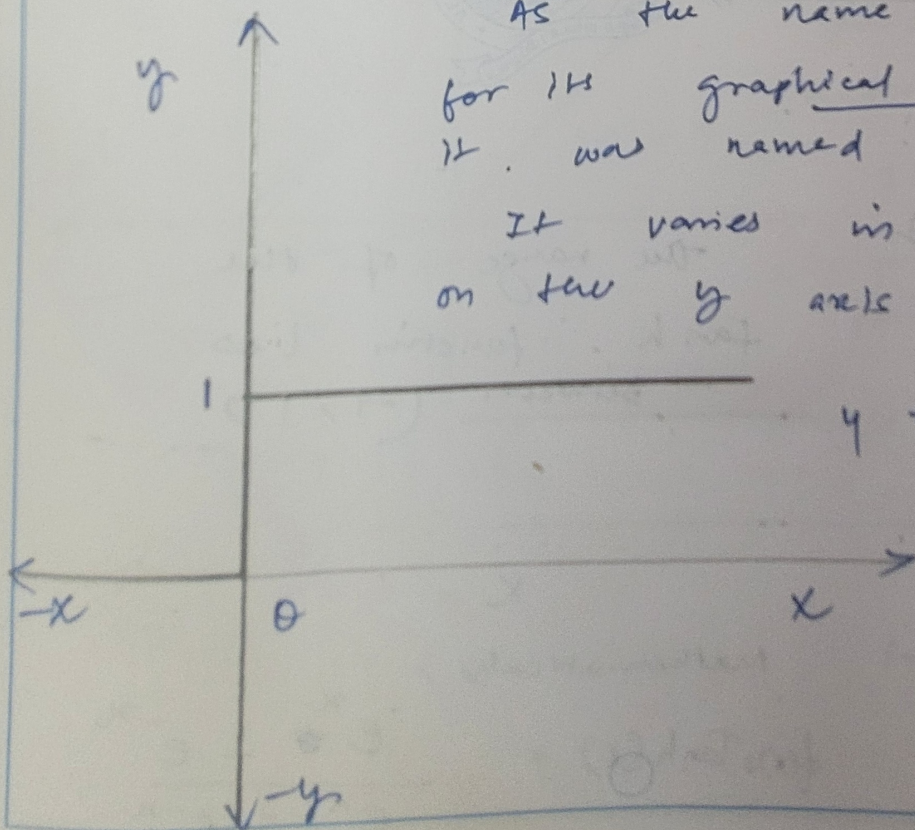
Here are some Activation functions, with their graph and function details.

① Step - Function :

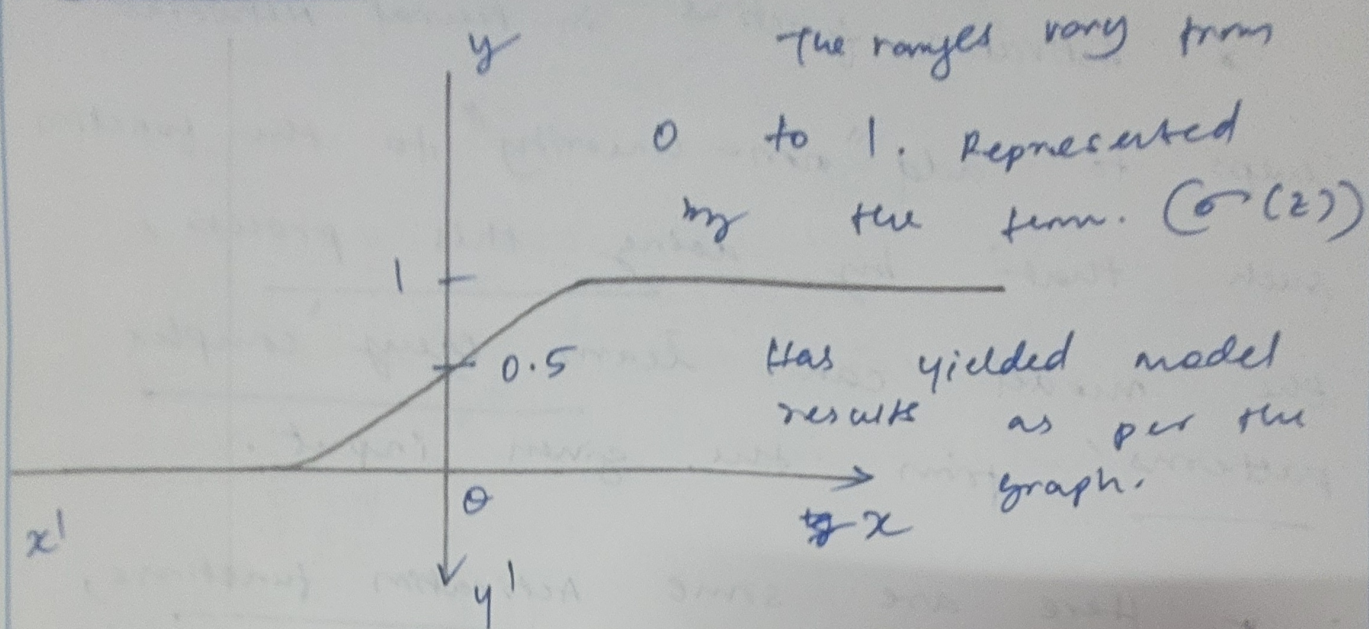
As the name suggests, for its graphical characteristics, it was named as step function.

It varies in the range on the y axis from 0 to 1.

$$y \in [0, 1]$$



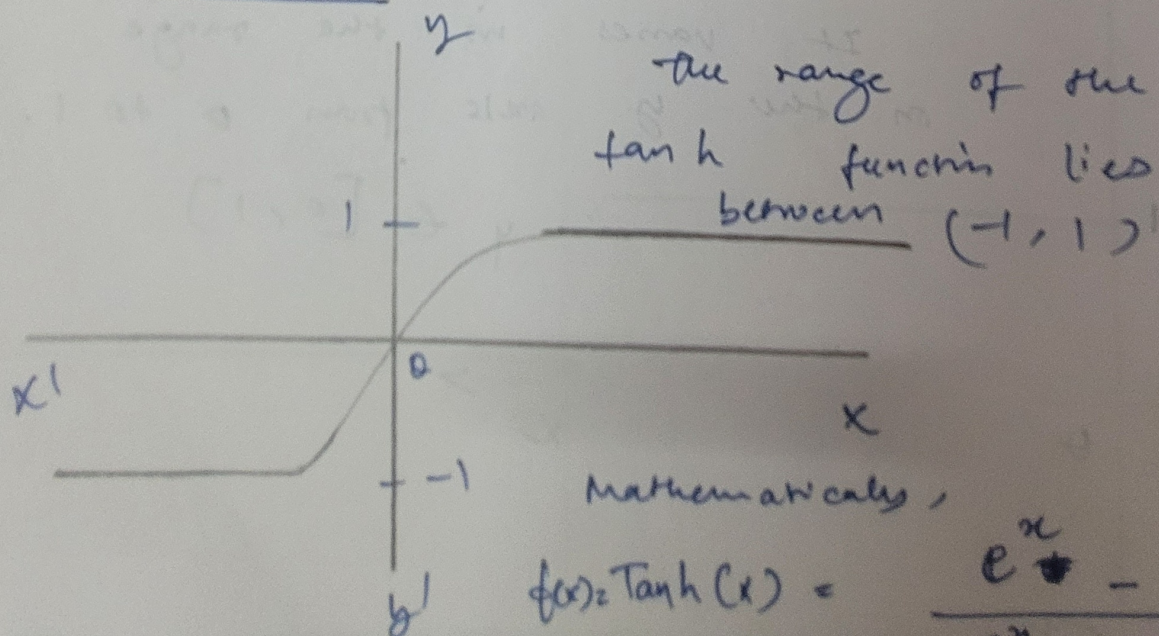
② Sigmoid Function :



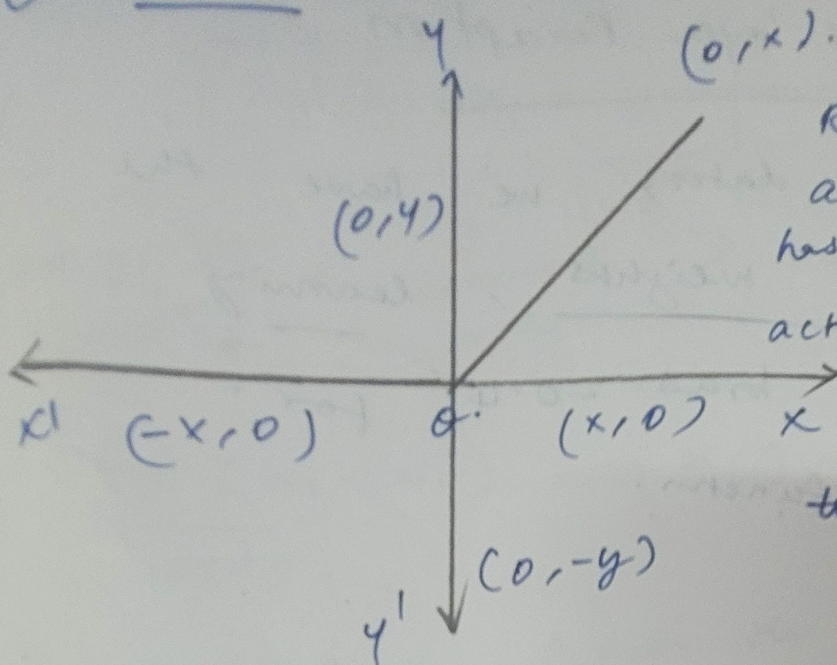
Mathematically, it is expressed as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

③ Tanh Function :



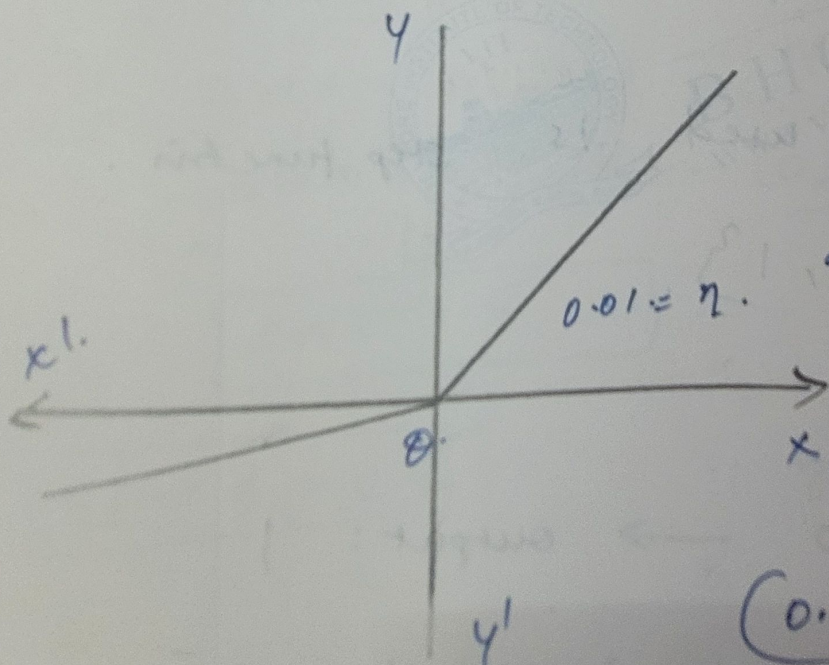
④ ReLU :



ReLU is one of the activation functions that has a 'steep-angled' activation function.

Helps identifying the complex patterns.

⑤ Leaky - ReLU :



Leaky ReLU is a variation of ReLU activation function.

It gives much diverse or varied pattern findings.

$(0.01x, x) \rightarrow$ leaky reLU.

5) To Design single layer Perceptron :

* From the given data, we have the details, of the weights, learning rate (α) and bias 20.4 for AND boolean function.

$$W_1 = 0.5$$

$$W_2 = 0$$

$$\text{bias} = 0.4$$

$$\alpha = 0.2$$

* Activation function used is step function.

$$\text{Output} = \{0, 1\}$$

* Range function:

$$f(x) \geq 0 \rightarrow \text{output: } 1$$

$$f(x) < 0 \rightarrow \text{output: } 0.$$

* Placing all the learn rules $\rightarrow$

Hebbian learning rule :

$$\Delta W = \eta \cdot x \cdot y \quad \text{output} \quad (1)$$

change in weights learning rate inputs

$$\frac{\partial (W_2 - W_1)}{\partial r \partial y} = \eta \cdot x \cdot y \quad (2)$$

$\eta \rightarrow \alpha$ (From the given question.)

$$\Delta W = W_2 - W_1 \quad \text{or} \quad \Delta W = W_{\text{final}} - W_{\text{initial}}$$

$$\Delta W = W_{\text{initial}} - W_{\text{final}}$$

$$\Delta W = [0.5] - [0]$$

$$\Delta W = 0.5 \quad (3)$$

$$\eta \rightarrow \alpha = 0.2$$

$$\therefore y = \sum_{i=1}^2 x_i w_i + \beta$$

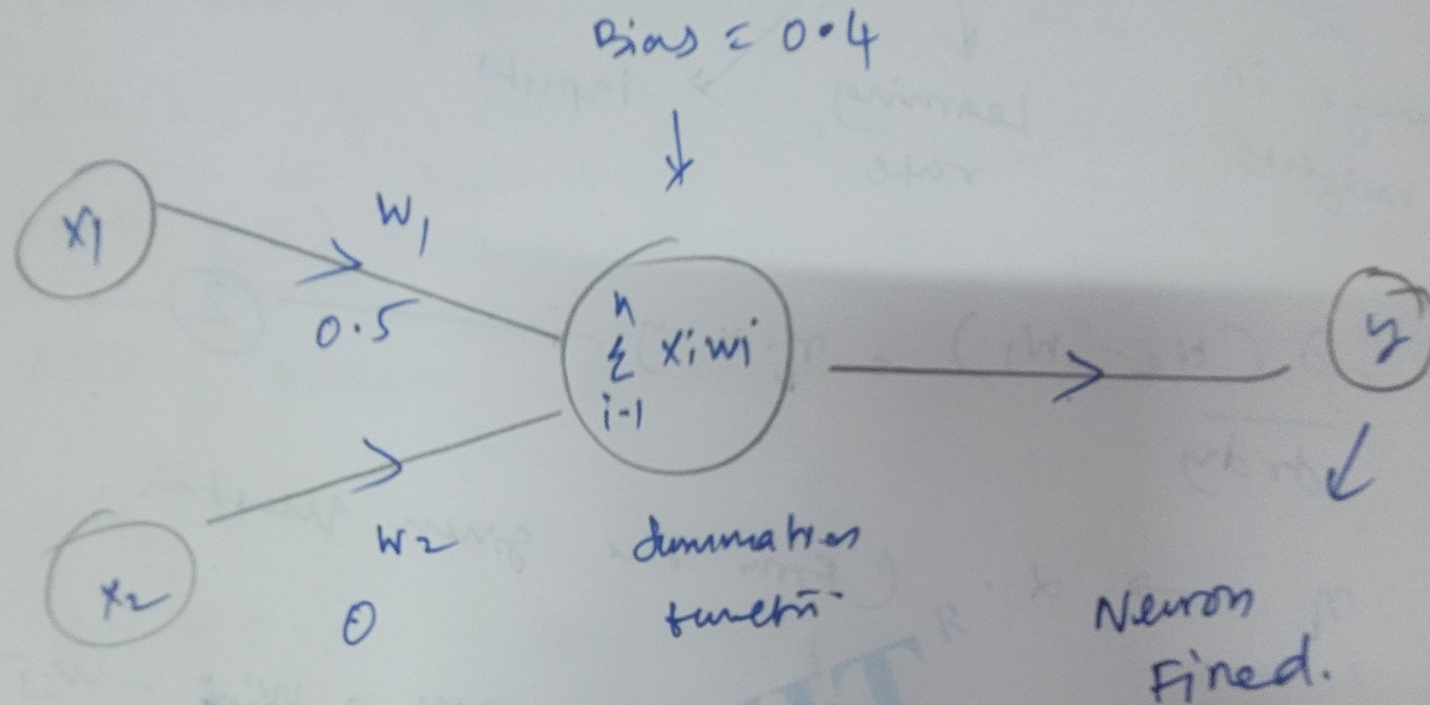
Now, applying in the formula.

$$0.5 = 0.2 \cdot x$$

$$x = \frac{0.5}{0.2}$$

$$x = 0.25$$

As, $f(x) > 0$, the output = 1.



$f(x) \geq 0 \rightarrow$ exceeded threshold value.